

Distal Vessel Occlusions in Acute Ischemic Stroke

Reframing Patient Selection and Reperfusion Strategies

Magda Jabłońska^{1,2} and Marc Ribó^{1,3}

Neurology® 2026;107:e218440. doi:10.1212/WNL.0000000000218440

Correspondence

Dr. Jabłońska
magda.jablonska@
icloud.com

Abstract

Mechanical thrombectomy (MT) has revolutionized the treatment of acute ischemic stroke because of large vessel occlusion (LVO), but distal and medium vessel occlusions (DMVOs) remain a gray zone. DMVOs account for a substantial proportion of ischemic strokes and can cause long-term disability or death despite often presenting with only mild to moderate symptoms. Their biologic and anatomic diversity—including small vessel caliber, variable perfusion territories, collateral dependence, and complex access pathways—creates distinct patterns of natural history, treatment responsiveness, and procedural risk compared with proximal LVO. Recent randomized trials and meta-analyses, conducted in broadly defined DMVO populations, did not demonstrate functional benefit of endovascular therapy (EVT) over best medical therapy (BMT) and observed numerically higher rates of intracranial hemorrhage with MT. These neutral results have fueled skepticism and guideline recommendations against routine EVT for DMVO stroke; however, they coexist with emerging trial data suggesting that treatment effects may depend on refined patient selection and optimized, distal-specific techniques, as well as with registry findings indicating that many patients with DMVO remain disabled despite BMT. In this narrative, we provide an interpretative synthesis of the available evidence to argue that the apparent discrepancy between biologic plausibility and neutral trial results reflects, at least in part, heterogeneity of DMVO definitions and phenotypes, limitations of severity and outcome scales, anatomy- and device-related safety constraints, and the unique mechanics and biology of small vessel thrombus retrieval rather than an absence of biologic efficacy. We highlight clinical, imaging, anatomic, and thrombus-related features which may help identify subgroups more likely to benefit from distal reperfusion and discuss emerging technical and pharmacologic strategies—including DMVO-specific device adaptations, intraarterial thrombolysis, and hybrid reperfusion approaches—which aim to reduce mechanical trauma while preserving the advantages of timely recanalization. Together, these considerations support a shift from viewing distal MT as a failed extension of proximal EVT to recognizing DMVO stroke as a distinct syndrome that requires tailored selection, techniques, and outcome assessment.

Introduction

Mechanical thrombectomy (MT) is a standard of care for acute ischemic stroke (AIS) caused by large vessel occlusion (LVO), with robust randomized evidence demonstrating efficacy across diverse clinical and imaging profiles.¹ In contrast, strokes due to distal and medium vessel occlusions (DMVOs)—commonly involving M2–M4 segments of the middle cerebral artery, anterior cerebral artery branches, and posterior cerebral artery territories—remain a gray zone.² DMVOs are estimated to account for 25%–40% of all ischemic strokes and can result in substantial long-term disability or death despite often presenting with mild to moderate neurologic deficits.³

Observational studies and real-world registries have suggested that endovascular treatment of DMVO can achieve high reperfusion rates with acceptable safety, fueling the expectation that MT might be extended to distal occlusions.⁴ However, 3 recent randomized controlled trials

¹Stroke Research, Vall d'Hebron Research Institute, Barcelona, Spain; ²2-nd Department of Radiology, Medical University of Gdansk, Poland; and ³Stroke Unit, Vall d'Hebron University Hospital, Barcelona, Spain.

Glossary

AIS = acute ischemic stroke; **BMT** = best medical therapy; **CT** = computed tomography; **DMVO** = distal and medium vessel occlusion; **DSA** = digital subtraction angiography; **EVT** = endovascular therapy; **IA** = intraarterial; **ISC** = International Stroke Conference; **IVT** = intravenous thrombolysis; **ivtPA** = intravenous tissue plasminogen activator; **LVO** = large vessel occlusion; **MCA** = middle cerebral artery; **MeVO** = medium vessel occlusion; **mRS** = modified Rankin Scale; **MT** = mechanical thrombectomy; **NIHSS** = National Institutes of Health Stroke Scale; **PCA** = posterior cerebral artery; **RCT** = randomized controlled trial; **SAH** = subarachnoid hemorrhage; **SAIL** = Stent Retriever–Assisted Intra-arterial Lysis; **sICH** = symptomatic intracranial hemorrhage; **SR** = stent retriever; **tPA** = tissue plasminogen activator.

challenged this assumption: ESCAPE-MeVO, DISTAL, and DISCOUNT did not show improved 90-day functional outcome with endovascular therapy compared with best medical therapy (BMT) and signaled a trend toward higher hemorrhagic complications.^{5,6} Recent meta-analyses have reported a significantly increased risk of symptomatic intracranial hemorrhage associated with endovascular therapy in DMVO populations, with estimates of up to a 2.5-fold increase.⁷ These findings have contributed to recent American Heart Association recommendations discouraging routine EVT for distal occlusions in current practice. At first glance, these neutral results might appear to close the chapter on distal MT. Yet, if we accept the axiom that timely recanalization and reperfusion should reduce final infarct volume, the key question becomes: why did this biologic premise not translate into functional benefit in randomized trials? Outcomes were poor in both study arms, with more than half of patients experiencing substantial disability or death at 90 days—underscoring an urgent necessity to identify strategies that can meaningfully improve prognosis in DMVO stroke. Importantly, emerging evidence suggests that treatment effects may be more likely to be observed when endovascular therapy is restricted to patients with moderate-to-severe neurologic deficits⁸ and when purpose-built, low-profile devices designed for distal vasculature are employed.⁹ However, these observations are not consistent across trials. Although the results of ORIENTAL-MeVO have been published, data from ongoing DISTALS trial remain preliminary and should be interpreted with caution.

Defining Distal and Medium Vessel Occlusions

A fundamental challenge in interpreting DMVO data is the absence of a unified and biologically grounded definition. Across the literature and clinical practice, medium and distal vessel occlusions have been variably defined, encompassing different arterial segments that differ substantially in vessel caliber, perfusion territory, collateral dependence, eloquence of the supplied cortex, and technical accessibility. As a result, occlusions grouped under the same label may represent biologically and clinically distinct entities.

Although anatomic definitions of distal vessel segments are generally well established, grouping different, well-defined occlusion types within a single analytical framework

complicates both trial interpretation and everyday decision-making regarding endovascular therapy. By aggregating heterogeneous occlusion types under broad anatomic categories, current classifications may obscure meaningful differences in natural history, treatment responsiveness, and hemorrhagic risk. Consequently, potential benefit in selected subgroups may be diluted when analyzed within broadly defined trial populations, limiting our ability to identify patients most likely to benefit from distal reperfusion strategies.

Clinical Phenotypes, Severity Assessment, and Patient Selection in DMVO Stroke

Clinical presentation of DMVO stroke spans a broad spectrum, and both extremes of severity pose challenges for identifying patients likely to benefit from endovascular reperfusion. Patients presenting with mild symptoms often have a favorable natural history, limited tissue at risk, or deficits that may improve spontaneously or with intravenous thrombolysis, constraining the measurable incremental benefit of invasive intervention. At the opposite extreme, very severe presentations may not reflect the distal occlusion itself but rather an earlier disease stage, such as a proximal LVO that has partially lysed or fragmented, with much of the ischemic injury already established.

Between these extremes lies a narrower sweet spot, potentially treatable clinical phenotype: patients with clearly disabling but not profound deficits, in whom symptoms are attributable to the distal occlusion itself and in whom salvageable tissue remains. Identifying this intermediate window may be more informative than relying on rigid severity thresholds when considering reperfusion strategies in DMVO stroke.

Accurate identification of this phenotype is complicated by limitations of commonly used global assessment tools, which may underestimate the functional relevance of nonmotor cortical deficits typical of DMVO stroke.

Age further modifies this clinical framework. With reduced neural reserve and plasticity, older patients often experience disproportionate functional consequences from relatively small infarcts. Accordingly, the relative impact of tissue injury caused

by a DMVO may be substantially greater in older individuals than in younger patients, challenging the use of uniform severity or infarct-size thresholds and underscoring the need to incorporate age into individualized selection strategies.¹⁰

Heterogeneity of Distal Occlusions

Distal occlusions are not clinically equivalent. An M3 occlusion supplying eloquent cortex in the dominant hemisphere may carry greater functional impact than a similar-sized occlusion in noneloquent territory, and an A3 occlusion in a young patient with minimal collaterals may behave very differently from a nondominant M2 branch in an elderly patient. Grouping such heterogeneous phenotypes under a single DMVO category and assessing recovery with a global disability scale such as mRS, dilutes any true treatment signal.¹¹ Although LVO trials benefited from relatively homogeneous proximal vascular territories, DMVO trials inherently encompass a wider spectrum of vessel calibers, perfusion territories, and collateral patterns, requiring more refined stratification than was implemented. Moreover, the inclusion of proximal or dominant M2 occlusions in some trials, and their exclusion in others, complicates pooling and can bias effect estimates toward the null, especially when more distal occlusions with smaller ischemic territories predominate.

Vascular Tortuosity, Caliber, and Access-Related Modifiers

Distal vessels are accessed through longer and more tortuous arterial pathways. Tortuosity of the aortic arch, cervical arteries, and intracranial segments can delay or prevent catheter access, increase procedural time, and impose mechanical stress on fragile distal vessels. Even advancement of a microwire or microcatheter may alter vessel geometry and tension on small branches and perforators. In this context, deployment and traction of stent retrievers designed for LVO can cause avulsion or perforation of small arteries, contributing to intracranial hemorrhagic complications.¹²

Clinical series have raised concern that the balance between benefit and harm may be less forgiving in medium and distal vessel MT, where procedural complications can disproportionately influence clinical outcome; in a large multicenter cohort, vessel perforation during DMVO MT was associated with 90-day mortality of ~50%, similar to outcomes after LVO perforation.¹³ Tortuosity, vessel caliber, and distance to the occlusion thus act as major determinants of safety, beyond the traditional concept of procedural difficulty. Quantitative assessment of vascular tortuosity may help identify patients in whom endovascular treatment is more likely to be successful, as well as those in whom the probability of procedural harm is unacceptably high.¹⁴ Such anatomy-based risk stratification was not systematically incorporated into patient selection in current DMVO trials.

Device-Vessel Mismatch

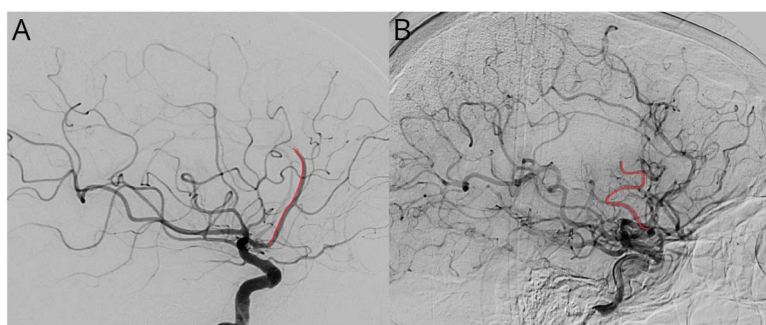
Most MT devices were developed and tested for proximal large vessels with diameters often exceeding 2.5–3 mm. When deployed in smaller distal arteries, their radial force, stiffness, and interaction with the vessel wall may be suboptimal, resulting in endothelial injury, dissection, vasospasm, perforation, and hemorrhagic transformation. This device-vessel mismatch is consistent with the observed signal of increased bleeding risk across DMVO trials, in which symptomatic intracranial hemorrhage (sICH) and any intracranial hemorrhage were more frequent with endovascular therapy. Although earlier randomized trials largely reflect experience with proximal MT devices adapted for distal use, emerging data from studies employing purpose-built, low-profile systems suggest that device optimization may play a role in revealing a potential clinical benefit of endovascular therapy in DMVO.⁹ Recent multicenter experience with distal-specific aspiration techniques is consistent with the concept that MT for DMVO represents a distinct technical entity requiring approaches tailored to distal vascular anatomy rather than extrapolated from proximal occlusion paradigms¹⁵ (Figure 1). However, available data remain preliminary and should be interpreted with caution because results from device-specific studies are not yet fully published.

Mechanics and Biology of Small Thrombus Retrieval

In distal occlusions, thrombi tend to be denser, more compact, and less compressible, often with reduced internal porosity and limited deformability. Thrombi are heterogeneous structures, typically composed of a fibrin- and platelet-rich outer shell surrounding a more deformable core. In distal vessels, this organization may be selectively enriched for the most rigid components.¹⁶ Indeed, distal occlusions may represent the residual, mechanically stiffer portion of a larger heterogeneous clot that has undergone partial lysis or mechanical fragmentation leaving behind the least deformable spongy parts. Histopathologic data further indicate that thrombi from medium and distal vessel occlusions include higher platelet and fibrin content than those from proximal occlusions, which may reduce susceptibility to purely mechanical retrieval.¹⁷

Biomechanical scaling laws further disadvantage small thrombi. At smaller length scales, higher relative forces are required to deform or mobilize a compact thrombus, while reductions in vessel radius amplify wall stress for a given applied force.¹⁸ This unfavorable scaling simultaneously increases the force needed for clot extraction and heightens the susceptibility of distal vessel walls to mechanical injury. Clinical studies show that platelet-poor thrombi and fewer retrieval attempts are independently associated with complete reperfusion,^{19,20} supporting a biologically plausible rationale for pharmacologic clot modification in DMVOs—such as

Figure 1 Digital Subtraction Angiography Illustrating Distal Medium Vessel Occlusions With Contrasting Distal Vascular Tortuosity



(A) Relatively straight distal M2 branch (low tortuosity). (B) Markedly tortuous distal M2 branch (high tortuosity).

adjunct intraarterial thrombolysis or targeted antiplatelet therapy—rather than simple escalation of mechanical force in small vessels. These considerations support the need for dedicated randomized trials evaluating intraarterial thrombolysis as both an adjunct and a potential alternative to mechanical thrombectomy in DMVO.

Are There Patients Who May Still Benefit?

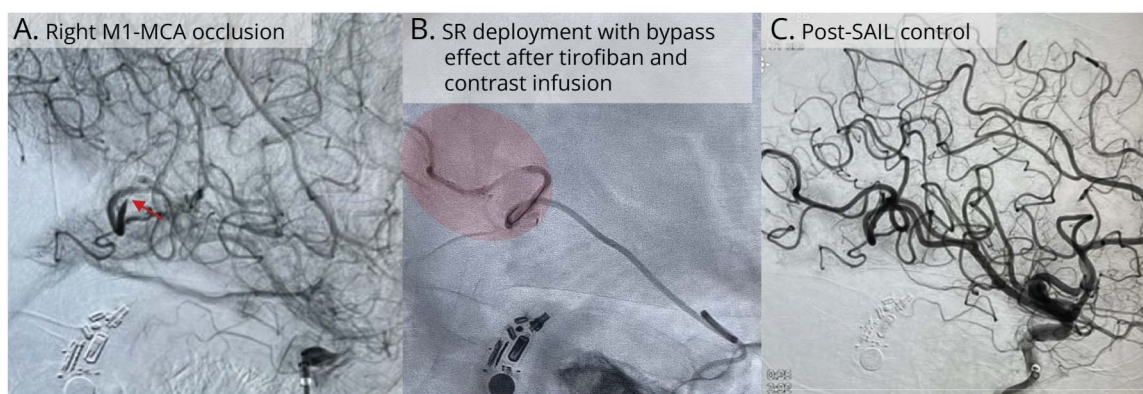
Despite neutral randomized trials, observational studies consistently demonstrate an association between successful reperfusion in DMVO and improved functional outcomes compared with unsuccessful reperfusion, with substantial absolute differences in rates of excellent and good outcome when reperfusion is achieved.³ Registry data suggest that up to half of patients with DMVO treated with BMT fail to achieve excellent functional outcome and roughly one-third remain functionally dependent at 90 days.³ This gap between natural history and the lack of benefit in aggregate trial populations

suggests that the right patients may not have been adequately targeted.²¹

Enrichment strategies combining clinical, imaging, and anatomic criteria may help identify subgroups with greater potential benefit of EVT. Patients in whom endovascular reperfusion may warrant consideration include those with higher NIHSS scores or clearly disabling eloquent deficits such as aphasia, hemianopia, or severe neglect, even when global NIHSS score is only moderate.²² Advanced imaging demonstrating substantial tissue at risk, small core, and favorable collaterals further supports potential benefit from reperfusion. Anatomic suitability—including relatively straightforward access, acceptable distal vessel caliber, and manageable tortuosity—may reduce procedural risk and increase the likelihood of achieving reperfusion without major complications.

Imaging-based analyses from recent DMVO trials provide important insights into potential strategies for improved patient selection, while also highlighting critical limitations of

Figure 2 Digital Subtraction Angiography of a SAIL Case in a Right M1 Middle Cerebral Artery Occlusion



(A) Baseline angiogram demonstrating thrombus in the right M1 segment (red arrow). (B) Stent retriever deployment with contrast and tirofiban infusion showing bypass effect at the occlusion site. (C) Post-SAIL control angiogram demonstrating complete reperfusion of the right MCA territory. MCA = middle cerebral artery; SAIL = Stent Retriever-Assisted Intra-arterial Lysis; SR = stent retriever.

Table Stroke-Related Symptoms Insufficiently Captured by the Modified Rankin Scale

Symptom/domain	Proposed assessment tool(s)
Global cognitive impairment	Montreal Cognitive Assessment (MoCA) Mini-Mental State Examination (MMSE)
Executive dysfunction	MoCA executive subtests; Trail Making Test (TMT B) Stroop Test
Attention and processing speed	Trail Making Test (TMT A) Digit Symbol Substitution Test
Memory impairment	MoCA memory subscale Rey Auditory Verbal Learning Test
Mild language impairment	NIHSS language item Boston Naming Test Token Test
Dysarthria	Frenchay Dysarthria Assessment Speech Intelligibility Rating
Visual field deficits	Automated perimetry; confrontation testing
Visuospatial dysfunction/neglect	Line Bisection Test Star Cancellation Test NIHSS neglect item
Fine motor impairment	Nine-Hole Peg Test Box and Block Test
Upper limb coordination	Fugl-Meyer Assessment (upper extremity)
Gait and balance	Timed Up and Go Test Berg Balance Scale
Physical endurance	6-Minute Walk Test
Sensory deficits	Fugl-Meyer sensory subscale; quantitative sensory testing
Central poststroke pain	Numeric Rating Scale DN4 questionnaire
Poststroke fatigue	Fatigue Severity Scale Multidimensional Fatigue Inventory
Depression	PHQ-9 Hamilton Depression Scale
Anxiety	GAD-7 Hospital Anxiety and Depression Scale
Emotional lability	Center for Neurologic Study–Lability Scale
Apathy	Apathy Evaluation Scale
Sleep disturbances	Pittsburgh Sleep Quality Index Epworth Sleepiness Scale
Health-related quality of life	Stroke Impact Scale EuroQol-5D
Participation and social reintegration	Reintegration to Normal Living Index
Return to work	Employment status Work Limitations Questionnaire

Several poststroke cognitive, neuropsychiatric, sensory, and participation-related deficits that substantially affect quality of life and functional reintegration are not adequately captured by the modified Rankin Scale and require complementary domain-specific assessment tools.

current paradigms. At the same time, caution is warranted when extrapolating LVO imaging criteria to distal occlusions. The presence of established deep infarction in the basal ganglia proximal to a DMVO should generally orient against endovascular intervention, as such injury may dominate

clinical outcome and limit the potential benefit of distal reperfusion. Importantly, perfusion imaging may underestimate irreversible injury in these regions: prior reperfusion of the infarcted basal ganglia can normalize perfusion parameters on CT perfusion despite established infarction,

leading to false-negative core assessment and an overestimation of salvageable tissue that will not benefit from a MT targeting a more distal occlusion.²³

Moreover, classical perfusion mismatch definitions validated in LVO—such as cerebral blood flow thresholds below 30% to define infarct core—may not be directly transferable to distal occlusions. In DMVO, ischemia often evolves under different hemodynamic conditions, with greater collateral influence, smaller territories, and distinct microvascular responses, suggesting that optimal perfusion thresholds for core and penumbra may differ from those established for proximal large vessels. Applying unadjusted LVO-based mismatch criteria to DMVO may therefore misclassify tissue viability and contribute to suboptimal patient selection.

Together, these observations argue for DMVO -specific imaging strategies that integrate, vascular tortuosity assessment, accurate infarct location (particularly deep vs cortical involvement), relative tissue-at-risk burden, collateral patterns, and adapted perfusion thresholds, rather than relying exclusively on paradigms developed for LVO.

These enrichment criteria are aligned with emerging expert consensus and sub-analyses of DMVO trials and do not contradict randomized evidence; rather, they suggest that initial trials may have preferentially tested distal MT in patients with limited room for improvement and higher susceptibility to treatment-related harm.

Beyond MT: Alternative and Adjunctive Strategies

Neutral DMVO trials underscore the need to reconsider not only whether endovascular reperfusion is performed in distal occlusions, but also how it is technically achieved. In small distal vessels, reducing mechanical stress while leveraging biological mechanisms of clot modification may be critical to improving safety and efficacy.

Intraarterial tissue plasminogen activator (tPA) delivered through low-profile microcatheters offers a plausible alternative or adjunct to MT in distal occlusions by enhancing fibrinolysis while minimizing direct mechanical trauma to fragile vessel walls. Early randomized and phase 2 studies—particularly protocols using adjunct intra-arterial tPA after successful or near-successful MT in proximal occlusions—suggest improved downstream microvascular reperfusion and higher rates of excellent functional outcome without significant hemorrhagic risk. However, these findings require confirmation and have not been specifically validated in primary DMVO populations.

Beyond pharmacologic therapy alone, technical modification of mechanical approaches is essential in distal vasculature. Full deployment and forceful retrieval of stent

retrievers—strategies developed for LVO—may be inappropriate in small distal arteries. Partial or controlled deployment of low-profile stent retrievers, with limited or no traction, can reduce radial and tensile forces on fragile vessel walls while allowing temporary bypass flow restoration and clot engagement.

These principles are integrated in pharmacologic-mechanical hybrid strategies such as Stent Retriever–Assisted Intraarterial Lysis (SAIL).²⁴ In SAIL, a partially deployed stent retriever functions as a temporary scaffold or bypass channel, facilitating targeted intraarterial delivery of thrombolytic or antiplatelet agents—most commonly alteplase or glycoprotein IIb/IIIa inhibitors—along the occluded or stenotic segment. The device is then resheathed or gently retrieved, minimizing endothelial injury, distal embolization, and hemorrhagic risk. Observational data suggest that SAIL can achieve sustained reperfusion in treatment-resistant distal occlusions and may obviate the need for angioplasty or permanent stenting in selected cases, although it requires advanced operator expertise and careful patient selection (Figure 2).

Taken together, these approaches emphasize that failure of conventional MT does not equate to failure of endovascular reperfusion in patients with DMVO. Rather, they point toward a paradigm in which technique adaptation, biologic clot modification, and minimization of mechanical trauma are central to safely treating distal occlusions. At present, these strategies remain supported primarily by observational and early-phase data and should be confined to experienced centers and rigorously designed studies, rather than used to justify routine distal MT outside controlled settings.

Clinical Implications

Current randomized evidence and meta-analyses do not support routine MT for all patients with distal and medium vessel occlusions. BMT, including intravenous tPA when eligible, remains appropriate for most patients, particularly those with mild deficits, small infarct cores, and limited tissue at risk.²⁵ In this context, neutral DMVO trials should not be interpreted as definitive futility of distal reperfusion, but rather as a call for more selective use of therapies optimized to the characteristics of the disease.

In carefully selected patients with clearly disabling symptoms, substantial salvageable tissue on imaging, and acceptable anatomic access, endovascular reperfusion may still be reasonable in experienced centers or within clinical trials, particularly for borderline anatomic locations situated between clearly indicated proximal occlusions and very distal ones. Conversely, for more distal M3/A3/P3 occlusions, current evidence favors a conservative approach, reserving MT or hybrid strategies for exceptional cases in which the expected disability is high, and the procedural risk can be minimized. Such

patient-level decision-making requires integration of clinical severity, vascular anatomy, thrombus biology, and local expertise.

An additional consideration in interpreting safety outcomes in DMVO trials relates to the definition of sICH. Current criteria of sICH were largely developed in the context of LVO, where hemorrhage volume is assessed relative to extensive infarct territories. In distal occlusions, however, the amount of tissue at risk is substantially smaller, such that even limited volumes of intracranial hemorrhage or focal subarachnoid hemorrhage may carry a disproportionate clinical impact. This scale mismatch raises the possibility that conventional sICH definitions may underestimate clinically meaningful hemorrhagic harm in DMVO stroke, particularly when small procedural hemorrhages affect eloquent cortex or occur in the setting of limited ischemic burden. Conversely, binary sICH classifications may fail to capture more subtle but functionally relevant bleeding patterns that influence outcome without meeting formal sICH thresholds. These considerations suggest that safety end points in DMVO trials may require recalibration, incorporating hemorrhage volume and location relative to the initial tissue at risk, rather than relying exclusively on definitions optimized for large infarcts.

Outcome Assessment Beyond the Modified Rankin Scale

Outcome assessment in DMVO also warrants reconsideration. The modified Rankin Scale is heavily weighted toward motor disability and dependency, yet distal occlusions frequently cause cognitive, visual, language, or neuropsychiatric deficits with minimal motor impairment. In addition, patient-perceived disability, including fatigue, anxiety, depression, and reduced quality of life, may be substantial and are often not captured by mRS (Table).

Future DMVO trials should incorporate patient-reported outcome measures and domain-specific assessments—such as detailed language, visuospatial, executive, and mood evaluations—to more accurately quantify benefit and harm. Imaging-based endpoints, including “tissue saved” on advanced imaging, may provide complementary information and correlate more closely with subtle deficits than global disability scales in small-territory strokes. Combining tissue-based metrics with patient-reported and domain-specific functional outcomes could yield more sensitive and mechanistically informative end points for distal reperfusion trials.

Future Directions

The future of distal reperfusion likely lies in redefining safety boundaries and optimizing biologic and anatomic selection rather than simply extending existing tools further into the vascular tree. Distal MT should not be viewed as

a miniaturized version of proximal MT; the technical skills, tactile feedback, and risk thresholds required for distal navigation differ fundamentally from those used in large vessels. It is plausible that the field advanced too rapidly toward efficacy trials without fully defining feasibility, safety limits, adequate tools, and training requirements.

Future studies should incorporate consecutive or minimally biased enrollment strategies, anatomy-based enrichment, and systematic assessment of vascular geometry—particularly supra-aortic access and intracranial vessel geometry—which may be facilitated by automated or AI-based tools.¹⁴ Such designs may also benefit from anatomically stratified randomization (e.g., based on M2 dominance) and direct comparison of reperfusion strategies—including intraarterial thrombolysis, mechanical thrombectomy, and best medical therapy—within more homogeneous subgroups. Device development must prioritize low-profile systems and techniques tailored to small vessels, with explicit caps on the number and force of retrieval attempts and predefined escalation pathways to pharmacologic or hybrid reperfusion strategies when mechanical risk becomes unacceptable. Adjunctive therapies targeting microvascular reperfusion and reperfusion injury, along with integration of tissue- and patient-centered end points, may further enhance net clinical benefit. Only through such integration of anatomy, biology, skills, and outcome assessment can the true therapeutic potential of distal reperfusion be determined.

Conclusion

In contrast to recent randomized trials reporting neutral results with broad inclusion criteria, the recently published results of the ORIENTAL-MeVO trial⁸ provide important additional context for interpreting these findings. Together with emerging data from the DISTALS trial,⁹ these results suggest that treatment benefit may depend on more stringent clinical selection and the use of devices specifically designed for distal occlusions. However, these findings should be interpreted with caution, as the full results of DISTALS have not yet been published, and the generalizability of these observations to broader MeVO populations remains to be established. These observations may reflect, at least in part, limitations in patient selection, anatomic heterogeneity, thrombus biology, and device-vessel mismatch rather than a complete absence of biologic efficacy. Routine distal MT for unselected DMVO stroke therefore cannot be recommended, and BMT—including ivtPA where appropriate—should remain the default strategy for most patients. At the same time, converging observational data, trial subanalyses, and mechanistic insights suggest that carefully selected patients, defined by disabling deficits, substantial salvageable tissue, favorable access anatomy, and appropriate thrombus characteristics, may still benefit from anatomically and biologically tailored reperfusion. Future research must integrate refined clinical and imaging selection, anatomy-aware device development,

operator training, and patient-centered outcome measures to define when, how, and for whom distal reperfusion can be transformed into a safe and clinically meaningful intervention.

Author Contributions

M. Jablonska: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

M. Ribo: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

Study Funding

The authors report no targeted funding.

Disclosure

The authors report no relevant disclosures. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

Publication History

Received by *Neurology*® March 20, 2026. Accepted in final form May 22, 2026. Submitted and externally peer reviewed. The handling editor was Editor-in-Chief José Merino, MD, MPhil, FAAN.

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